



Original Article

Explainable machine learning for predicting aspiration pneumonia after radiotherapy in oral cavity cancer

Yu-Fan Wu^a, Jhen-Bin Lin^b, Yi-Shing Leu^c, Fang-Ju Sun^{d,e}, Yu-Jen Chen^f, Jie Lee^{f,g,*}

^a Department of Information Management, Fu-Jen Catholic University, New Taipei City, Taiwan

^b Department of Radiation Oncology, Changhua Christian Hospital, Changhua, Taiwan

^c Department of Otolaryngology and Head Neck Surgery, MacKay Memorial Hospital, Taipei, Taiwan

^d Department of Medical Research, MacKay Memorial Hospital, Taipei, Taiwan

^e Institute of Biomedical Informatics, National Yang Ming Chiao Tung University, Taipei, Taiwan

^f Department of Radiation Oncology, MacKay Memorial Hospital, Taipei, Taiwan

^g Department of Medicine, MacKay Medical University, New Taipei City, Taiwan

ARTICLE INFO

Keywords:

Aspiration pneumonia
Explainable artificial intelligence
Machine learning
Oral cavity cancer
Radiotherapy

ABSTRACT

Background and purpose: Aspiration pneumonia is a severe late event that occurs following radiotherapy for oral cavity cancer (OCC). This study aimed to develop explainable machine learning models to predict aspiration pneumonia.

Materials and methods: This study included 880 patients with OCC (derivation cohort, 561 patients; external validation cohort, 319 patients) who underwent surgery and adjuvant radiotherapy between 2010 and 2020. Random Forest (RF), eXtreme Gradient Boosting (XGBoost), and Categorical Boosting (CatBoost) models were trained to predict aspiration pneumonia using clinical and dosimetric features. Model performance was evaluated using the area under the curve (AUC). The SHapley Additive exPlanations (SHAP) method was applied to rank feature importance and explain the model.

Results: The RF model achieved the highest AUC compared with the XGBoost and CatBoost models in the external validation set (0.966, 0.962, and 0.955, respectively). An explainable RF model identified the mean doses to the superior pharyngeal constrictor muscle (PCM), middle PCM, and supraglottic larynx as the three most important features for predicting aspiration pneumonia in the training and validation sets. The dose–toxicity relationship between the mean dose to the swallowing structures and aspiration pneumonia was nonlinear. The threshold mean dose to the swallowing structures for lowering the risk of aspiration pneumonia was determined. The SHAP force plot provided an individualized interpretation of the model predictions for each patient.

Conclusion: An explainable machine learning model can assist clinicians in predicting aspiration pneumonia after radiotherapy for OCC. Individualized predictions may help tailor radiotherapy to reduce the risk of aspiration pneumonia.

Introduction

Oral cavity cancer (OCC) is the most common head and neck malignancy regarding incidence and mortality, with an estimated 389,485 new cases and 188,230 deaths worldwide by 2022 [1]. Surgery is the cornerstone of OCC treatment and adjuvant radiotherapy can reduce locoregional recurrence and improve survival outcomes in patients with pathological risk factors [2–4]. However, radiotherapy for OCC can induce severe side effects that can significantly affect survival. Aspiration pneumonia is a potentially life-threatening late adverse event and a

major cause of noncancer-related mortality following radiotherapy for OCC [5–12]. Therefore, there is a clear clinical need for a method to predict aspiration pneumonia and potentially guide the modification of risk factors to prevent aspiration pneumonia.

Machine learning (ML) algorithms have shown favorable performance in predicting the outcomes of head and neck radiotherapy [13]. ML algorithms that can handle complex interactions and nonlinear relationships could favorably predict aspiration pneumonia using clinical and dosimetric factors. With the advancement of explainable artificial intelligence, the “black box” issue of ML can be addressed, thereby

* Corresponding author at: Department of Radiation Oncology, MacKay Memorial Hospital 92, Section 2, Chung Shan North Road, Taipei 104217, Taiwan.
E-mail address: sinus.5706@mmh.org.tw (J. Lee).

<https://doi.org/10.1016/j.radonc.2025.111066>

Received 13 May 2025; Accepted 28 July 2025

Available online 30 July 2025

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Table 1
Patient characteristics.

Characteristics	Training set (n = 393)	Internal validation set (n = 168)	External validation set (n = 319)	p
Age (years)	55 (48–63)	55 (46–61)	56 (48–64)	0.23
Sex				0.85
Female	56 (14.2)	21 (12.5)	45 (14.1)	
Male	337 (85.8)	147 (87.5)	274 (85.9)	
CCI	1 (1–2)	1 (0–2)	1 (1–2)	0.09
SMI	50.7 (45.6–55.4)	51.5 (45.6–55.5)	50.7 (45.4–55.2)	0.46
Tumor site				0.46
Tongue	132 (33.6)	68 (40.5)	106 (33.2)	
Buccal mucosa	170 (43.3)	54 (32.1)	118 (37.0)	
Other subsites	91 (23.1)	46 (27.4)	95 (29.8)	
Stage				0.23
I–II	58 (14.8)	31 (18.5)	62 (19.4)	
III–IVB	335 (85.2)	137 (81.5)	257 (80.6)	
Smoking status				0.73
Nonsmoker	133 (33.8)	60 (35.7)	117 (36.7)	
Smoker	260 (66.2)	108 (64.3)	202 (63.3)	
Adjuvant treatment				0.18
Radiotherapy	142 (36.1)	71 (42.3)	135 (42.3)	
Chemoradiotherapy	251 (63.9)	97 (57.7)	184 (57.7)	
Mean dose to organs				
Superior PCM	50.8 (48.1–53.0)	50.4 (48.1–52.9)	50.8 (46.5–56.1)	0.79
Middle PCM	47.4 (45.1–49.5)	47.8 (45.9–49.5)	47.0 (44.1–50.3)	0.18
Inferior PCM	43.0 (40.6–45.8)	43.7 (41.5–46.1)	42.2 (38.9–45.9)	0.06
Cricopharyngeus muscle	39.3 (35.4–42.8)	38.9 (35.0–43.3)	40.3 (33.9–44.6)	0.67
Esophagus inlet muscles	37.1 (32.9–41.4)	37.1 (32.5–42.0)	37.4 (30.9–43.5)	0.94
Cervical esophagus	36.1 (31.8–40.5)	35.6 (30.9–40.3)	36.9 (30.5–41.9)	0.18
Supraglottic larynx	44.8 (42.0–46.8)	45.3 (42.3–47.5)	44.6 (41.9–48.1)	0.54
Glottic larynx	42.0 (39.1–43.9)	42.3 (40.0–44.8)	42.2 (37.8–45.6)	0.40

Abbreviations: CCI, Charlson Comorbidity Index; PCM, pharyngeal constrictor muscle; SMI, skeletal muscle index.
Data are median (interquartile range) or number (%).

facilitating its utility in clinical practice. The SHapley Additive ExPlanations (SHAP) method can visualize the inner workings of an ML model and provide interpretations at the cohort or patient level [14–16]. Through individualized predictions using the SHAP method, important modifiable factors may be identified, providing strategies for lowering the risk of aspiration pneumonia. A previous study successfully developed an explainable ML model to predict the need for enteral nutrition during radiotherapy for head and neck cancer [17]; however, scarce research has used explainable ML to predict aspiration pneumonia after radiotherapy for OCC.

The radiation dose delivered to the swallowing structures is associated with the risk of aspiration in head and neck cancers; however, dose–toxicity relationships are potentially nonlinear and involve complex interactions. Furthermore, common statistical methods, such as linear and logistic regression, might not optimally characterize them. ML and SHAP methods allow for the modeling of nonlinear relationships and complex interactions and may characterize the dose–toxicity relationship [18], providing information that could be used to optimize radiotherapy plans.

In this study, we aimed to develop and validate an explainable ML model to predict aspiration pneumonia following radiotherapy in patients with OCC.

Methods

Patients

This study was approved by the Institutional Review Boards of our institutions, and the requirement for informed consent was waived due to the retrospective nature of this study. This study adhered to the TRIPOD AI statement to develop and report our study (Supplementary material) [19].

Patients newly diagnosed with OCC treated with surgery and adjuvant radiotherapy at Changhua Christian Hospital (derivation cohort, *n* = 748) and MacKay Memorial Hospital (external validation cohort, *n* = 491) between 2010 and 2020 were reviewed. Patients were excluded if they had (i) a history of malignancy, (ii) insufficient clinical data, (iii) insufficient radiotherapy data, (iv) pneumonia before radiotherapy, or (v) missing or insufficient follow-up data. Based on these criteria, 561 patients in the derivation cohort and 319 patients in the external validation cohort were included in the analysis (Supplementary Fig. 1).

All patients underwent radical tumor excision surgery and neck dissection. Adjuvant radiotherapy was administered with intensity-modulated radiotherapy (IMRT) with a simultaneous integrated boost technique. The total dose was 60–66 Gy to the tumor bed and dissected nodal region and 54–56 Gy to the elective region [20–22]. The addition of cisplatin chemotherapy to radiotherapy was determined based on major risk factors by multidisciplinary teams.

Outcome measures

The primary outcome measure was aspiration pneumonia. Patients were considered to have aspiration pneumonia if they met the following criteria: (i) clinical symptoms with corresponding characteristic radiographic changes, and (ii) the presence of aspiration previously identified on video fluoroscopy or fiberoptic endoscopic evaluation of swallowing or suspected clinically (choking or delayed swallowing). Data on the date and occurrence of aspiration pneumonia and the date of the last follow-up were extracted from the institutional database.

The following clinical data were collected from the institutional database: age, sex, Charlson Comorbidity Index (CCI), skeletal muscle index (SMI), primary cancer site, stage, smoking status, and chemotherapy. The swallowing structures, including the superior pharyngeal constrictor muscle (SPCM), middle PCM (MPCM), inferior PCM (IPCM), cricopharyngeus muscle, supraglottic larynx, glottic larynx, esophagus inlet muscle, and cervical esophagus, were delineated by a radiation oncologist according to the guidelines [23]. The mean dose data for each swallowing structure were derived from the original radiotherapy plan, based on previous research [6–11]. Spearman correlation coefficient was applied to assess correlations between mean doses to the swallowing structures, using a threshold of 0.8 for high multicollinearity [24]. All Spearman correlation coefficients were below the threshold, and no dosimetric features were excluded (Supplementary Table 1).

Machine learning models

An overview of the study design is shown in Supplementary Fig. 1. Data from the derivation cohort were divided, with 70 % used for training and 30 % used for validation (internal validation). An external dataset was used for testing (external validation).

Potential features were preselected according to previous research: age, sex, CCI, SMI, primary cancer site, stage, smoking status, chemotherapy, and mean dose to the swallowing structures [5–12]. The Random Forest (RF), eXtreme Gradient Boosting (XGBoost), and Categorical Boosting (CatBoost) models were trained using the Python Scikit-learn library (version 1.4.0; <https://github.com/scikit-learn/scikit-learn.git>), XGBoost (version 2.0.3; <https://github.com/dmlc/xgboost.git>), and CatBoost (version 1.2.2; <https://github.com/catboost/catboost.git>) [25]. All model hyperparameters were optimized using Optuna, a

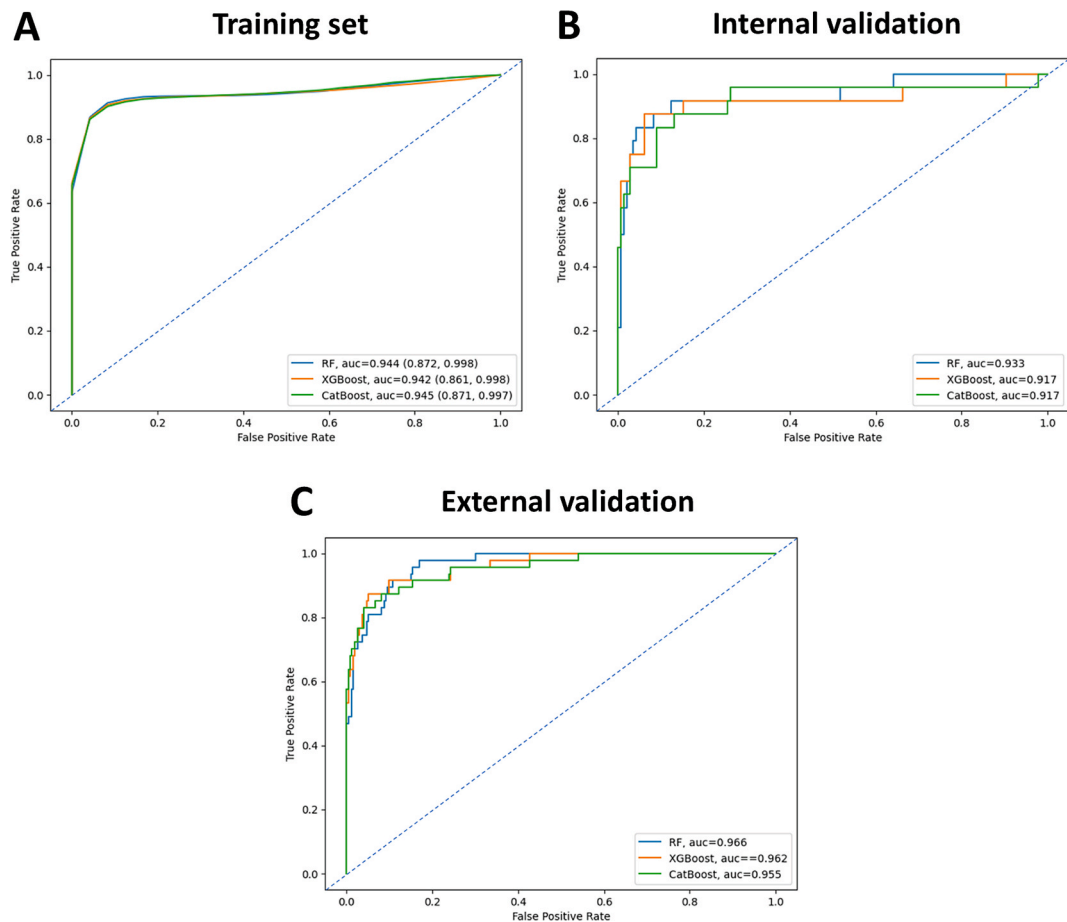


Fig. 1. Receiver operating characteristic curves showing the aspiration pneumonia predictive performance of the machine learning models in the (A) training set, (B) internal validation sets, and (C) external validation set. CatBoost, Categorical Boosting; RF, Random Forest; XGBoost, eXtreme Gradient Boosting.

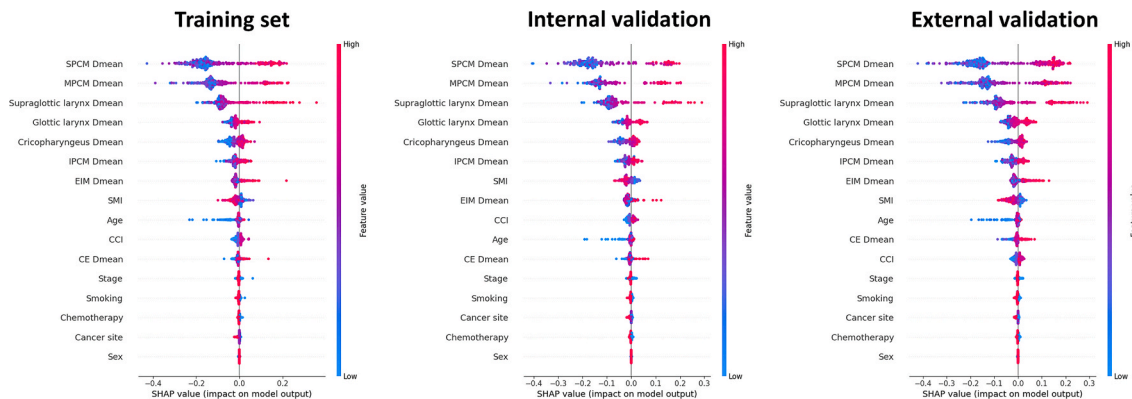


Fig. 2. SHAP summary plots of Random Forest model in the training and validation sets. SHAP summary plots rank the features in order of the global importance in the prediction. Each dot represents a SHAP value for a feature per patient, and red to blue colors represent the feature's value from high to low, respectively. SHAP, SHapley Additive exPlanations. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Python implementation of Bayesian optimization, to maximize the area under the curve (AUC). A borderline synthetic minority oversampling technique was applied to address the effects of imbalanced data and to produce a robust model [26]. Model performance was evaluated using the AUC, F1-score, sensitivity, specificity, and precision. These metrics were assessed using 500 bootstrap resamples from the training set, and the results were reported with 95 % confidence intervals (CIs). The final model was selected based on its performance in the internal and external validation cohorts and subsequently analyzed using SHAP to interpret the prediction of aspiration pneumonia.

Shapley additive exPlanations

The SHAP values were calculated to evaluate the contribution of each input feature to the prediction of aspiration pneumonia, with values of 0 equivalent to a neutral effect, values >0 equivalent to a positive effect (i.e., a higher risk of aspiration pneumonia), and vice versa. The SHAP summary plots illustrate the importance of the features within the model. These plots provide a global overview of the features that influence the ML models. Dependence plots were used to visualize the relationships between features and aspiration pneumonia. In the

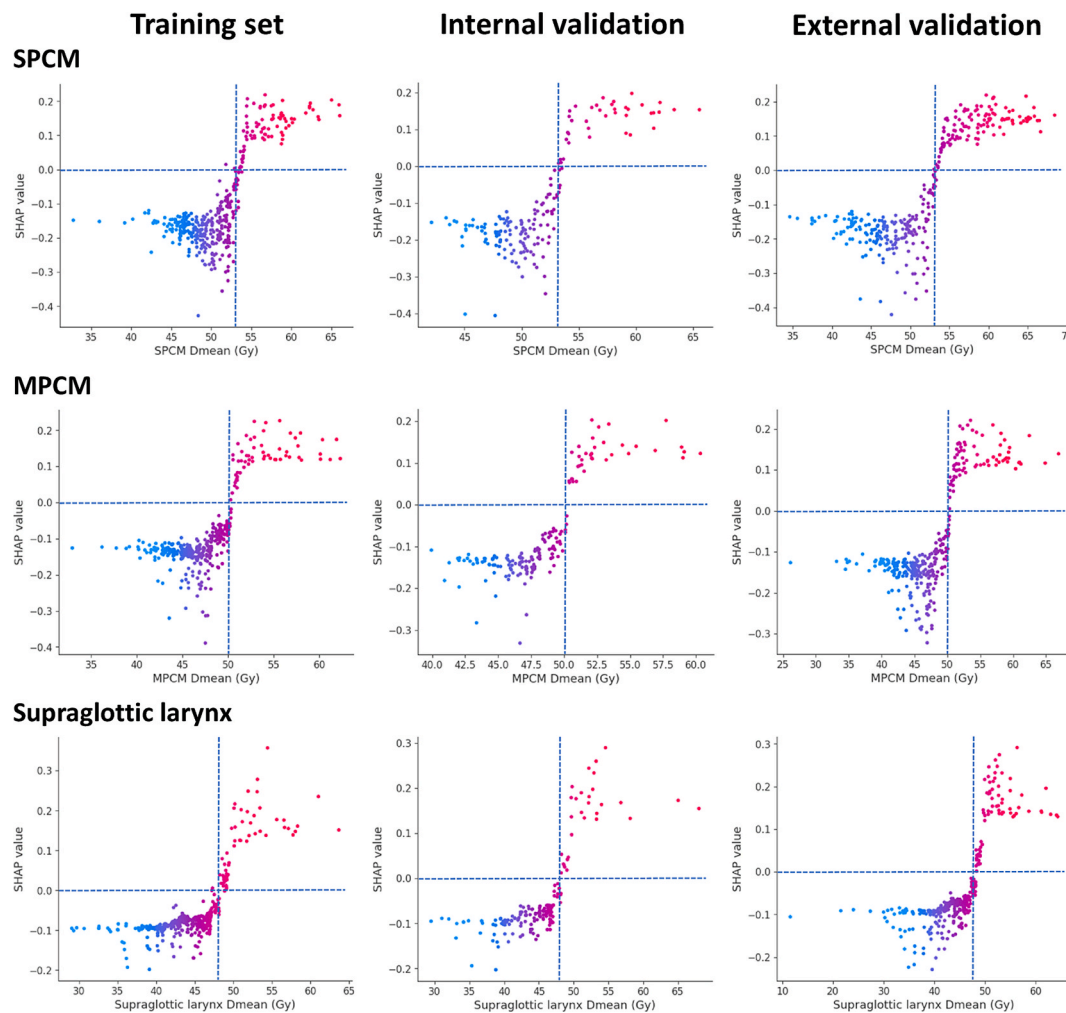


Fig. 3. SHAP dependence plots of SPCM, MPCM, and supraglottic larynx in the derivation and validation sets. Each dot represents a SHAP value for a feature per patient, and red to blue colors represent the feature's value from high to low, respectively. SHAP values for specific features exceeding zero represent an increased risk of aspiration pneumonia. MPCM, middle pharyngeal constrictor muscle; SHAP, SHapley Additive exPlanations; SPCM, superior pharyngeal constrictor muscle. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

SHAP dependence plots, the feature values are plotted on the x-axis, and the SHAP values are plotted on the y-axis. The dosimetric threshold could be identified by focusing on where the SHAP values exceeded zero, which pushed the prediction toward the aspiration pneumonia class.

Force plots were created to interpret the contributions of the features to the predictions at the patient level. SHapley Additive ExPlanations version 0.40.0 was used for the analysis.

Statistical analyses

Continuous data were presented as mean and standard deviation or median and interquartile range (IQR) and was compared using analyses of variance or the Kruskal–Wallis rank sum test, as statistically appropriate. Categorical data were presented as frequencies and percentages and compared using the chi-square or Fisher's exact test for categorical variables. The Kaplan–Meier method was used to estimate the cumulative incidence of aspiration pneumonia. All analyses were performed using Python version 3.7.3 and IBM SPSS version 26.0 (IBM Corp., Armonk, NY, USA). Statistical significance was set at $p < 0.05$ (two-sided).

Results

A total of 880 patients were included in the analysis (derivation cohort: 561 patients; external validation cohort: 319 patients). The 561 patients in the derivation cohort were allocated to separate training ($n = 393$) and internal validation ($n = 168$) sets. The patient characteristics are presented in Table 1. No significant differences in patient characteristics were observed between the training, internal validation, and external validation sets.

With a median follow-up of 5.8 (IQR, 2.9–9.4), 5.6 (IQR, 3.2–10.7), and 4.7 (IQR, 2.8–8.1) years, 51 (13.0 %), 21 (12.5 %), and 47 (14.7 %) patients had aspiration pneumonia in the training, internal validation, and external validation sets, respectively. The median time to aspiration pneumonia after radiotherapy was 11.1 (IQR: 7.3–16.2), 13.5 (IQR: 7.5–22.7), and 10.3 (IQR: 3.0–22.5) months, respectively. The 5-year cumulative incidences of aspiration pneumonia were 13.4 %, 13.2 %, and 15.9 %, respectively. Patients with aspiration pneumonia were more likely to have higher age, CCI, and mean dose to the SPCM, MPCM, IPCM, cricopharyngeus muscle, esophagus inlet muscles, supraglottic larynx, and glottic larynx, and a lower SMI (Supplementary Table 2).

In the training set, the RF, XGBoost, and CatBoost models achieved favorable performances, with AUCs of 0.944 (95 % CI: 0.872–0.998), 0.942 (95 % CI: 0.861–0.998), and 0.945 (95 % CI: 0.871–0.997), respectively (Fig. 1, Supplementary Table 3). The hyperparameters are

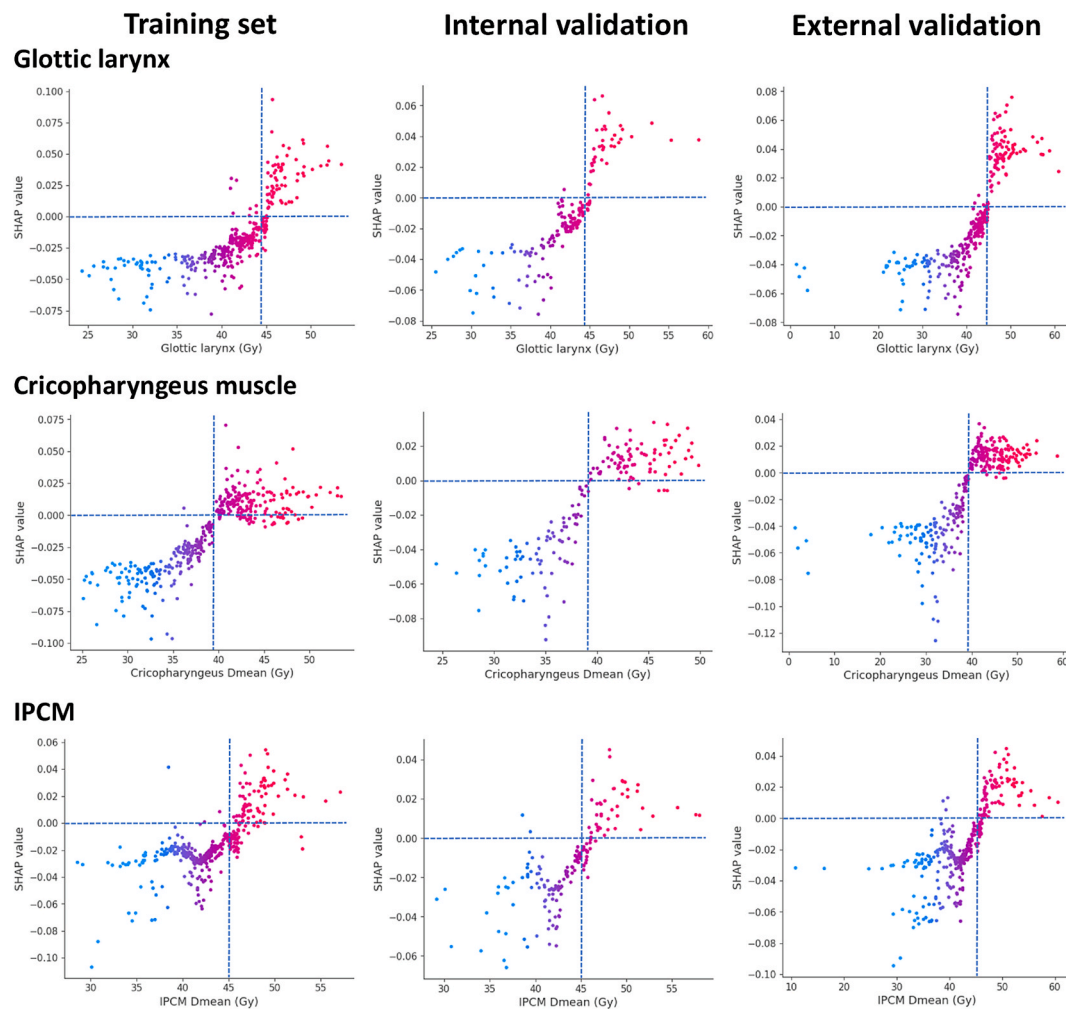


Fig. 4. SHAP dependence plots of glottic larynx, cricopharyngeus muscle, and IPCM in the derivation and validation sets. Each dot represents a SHAP value for a feature per patient, and red to blue colors represent the feature's value from high to low, respectively. SHAP values for specific features exceeding zero represent an increased risk of aspiration pneumonia. IPCM, inferior pharyngeal constrictor muscle; SHAP, SHapley Additive exPlanations. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

listed in [Supplementary Table 4](#). In the internal validation cohort, the RF model achieved the highest AUC of 0.933, whereas the XGBoost and CatBoost models achieved AUCs values of 0.917 and 0.917, respectively. In the external validation cohort, the RF model achieved the highest AUC of 0.966, whereas the XGBoost and CatBoost models achieved AUCs values of 0.962 and 0.955, respectively. Based on these results, the RF model was selected for the SHAP analysis.

The SHAP summary plots provided a global explanation of the RF model and ranked the importance of each feature in predicting aspiration pneumonia ([Fig. 2](#)). In the training and validation sets, the mean doses to the SPCM, MPCM, and supraglottic larynx were the top three important features for predicting aspiration pneumonia, followed by the glottis larynx, cricopharyngeus muscle, and IPCM. The mean doses to the esophagus inlet muscles and the cervical esophagus contributed less to the prediction.

The SHAP dependence plots depicted positive nonlinear relationships between the mean dose to the swallowing structures and the SHAP value, suggesting that an increase in the mean dose to these swallowing structures is associated with a higher risk of aspiration pneumonia ([Figs. 3 and 4](#)). By inspecting where the SHAP values exceeded 0, the dosimetric thresholds predictive of aspiration pneumonia could be identified. In the training and validation sets, the thresholds of the mean dose to predict aspiration pneumonia were 53.0 Gy to the SPCM, 50.0 Gy to the MPCM, 48.0 Gy to the supraglottic larynx, 44.5 Gy to the glottis

larynx, 39.0 Gy to the cricopharyngeus muscle, and 45 Gy to the IPCM.

The SHAP force plots provided an individualized prediction of aspiration pneumonia at the patient level. [Fig. 5](#) shows the clinical and dosimetric features of two patients. Patient A developed aspiration pneumonia after radiotherapy, and the RF model predicted an 88 % probability of aspiration pneumonia. The mean dose to the SPCM (56.9 Gy), MPCM (51.8 Gy), supraglottic larynx (56.1 Gy), IPCM (51.9 Gy), and glottic larynx (46.9 Gy), and age were the main predictors of aspiration pneumonia. Patient B did not have aspiration pneumonia; the RF model predicted an 11 % probability of aspiration pneumonia. These results suggest that the mean dose to the SPCM (48.0 Gy), MPCM (46.0 Gy), supraglottic larynx (46.3 Gy), esophageal inlet muscle (42.0 Gy), and IPCM (45.9 Gy) were the main contributors for this prediction.

Discussion

We developed an ML model to predict aspiration pneumonia in patients with OCC. To the best of our knowledge, this is the first study to use a cohort consisting of only patients with OCC for model development. The model was trained on clinical and dosimetric data that are easily available in clinical practice to provide predictions. The RF model exhibited the best performance and was validated using external data. The relationship between the mean dose to the swallowing structures and aspiration pneumonia was nonlinear. The SHAP method provided

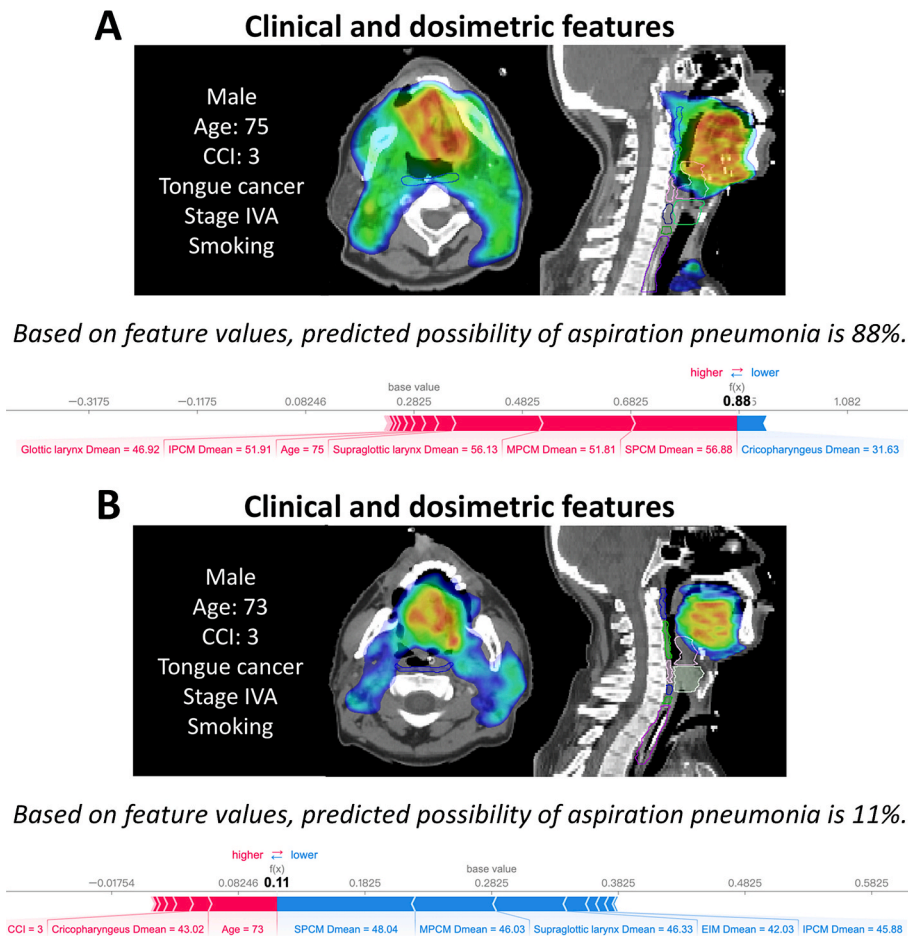


Fig. 5. SHAP force plots of (A) patient with aspiration pneumonia and (B) patient without aspiration pneumonia. The color represents the contributions of each feature, with red being positive and blue being negative. The length of the color bar represents the contribution strength. SHAP, SHapley Additive exPlanations. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

insightful interpretations of the model at the global and patient levels.

Laryngeal vestibular closure, hyolaryngeal elevation, tongue base retraction, and pharyngeal contraction are the most prone to radiation damage and are correlated with aspiration [10]. The supraglottic larynx, PCM, and functional swallowing units, including the anterior segment and the hyoglossus/styloglossus complex, are crucial aspiration-related structures. A previous study developed normal tissue complication probability (NTCP) models to predict aspiration based on the mean dose to aspiration-related structures in patients with pharyngeal/laryngeal cancers undergoing definitive radiotherapy [9–11]. In this study, we only included patients with OCC who underwent surgery and adjuvant radiotherapy because the treatment and target volume design of OCC can differ from those of pharyngeal/laryngeal cancer. Notably, the anterior segment and hyoglossus/styloglossus complex were not analyzed because these structures are likely to be mobilized or dissected during surgery for OCC. Therefore, a specific model for aspiration pneumonia in OCC has been suggested.

The ML model in this study showed favorable performance in predicting aspiration pneumonia. In descending order, the mean doses to the SPCM, MPCM, supraglottic larynx, glottic larynx, cricopharyngeus muscle, and IPCM were the six most important features for predicting aspiration pneumonia. A possible explanation is that these swallowing structures are relevant to laryngeal vestibular closure and pharyngeal contraction and that radiation damage to these structures is associated with the risk of aspiration [10]. The SHAP method also revealed a nonlinear dose–toxicity relationship for aspiration pneumonia and identified possible dose thresholds associated with aspiration pneumonia. However, we suggest inputting the mean doses of these

swallowing structures and clinical factors into an ML model, which can handle complex interactions and nonlinear relationships to predict aspiration pneumonia. Using this approach, these factors can be comprehensively evaluated for prediction. For instance, patient A (Fig. 5) had a higher risk of aspiration pneumonia, which was mainly attributed to the higher mean doses to the swallowing structures. This individualized prediction may be tailored to optimize the radiotherapy plan to lower the mean dose to the swallowing structures and the risk of aspiration pneumonia.

This study had some limitations. First, although our ML model was externally validated, further validation of the model in a more diverse population is required. Second, several features were potentially predictive of aspiration pneumonia, including pretreatment swallowing function, dysphagia, and smoking intensity, as most of this information was available for only a few patients and could not be analyzed. Third, aspiration pneumonia may have been under-recognized by clinicians and under-reported by patients in this study. Despite these limitations, this is the largest study to develop an ML model for predicting aspiration pneumonia in patients with OCC. The model was developed using clinical and dosimetric data that can be accessed in clinical practice and was externally validated.

Conclusions

We constructed and validated an ML model to predict aspiration pneumonia following radiotherapy for OCC. The model was based on accessible clinical and dosimetric data in clinical practice and had a favorable prediction performance. The SHAP method provided an

interpretation of predictions and individualized insights into the important factors associated with aspiration pneumonia. These approaches could provide information to identify patients who are at risk of aspiration pneumonia and guide management strategies to lower the risk and improve the overall care for these patients.

CRediT authorship contribution statement

Yu-Fan Wu: Formal analysis, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Jhen-Bin Lin:** Investigation, Methodology, Writing – review & editing, Data curation, Formal analysis. **Yi-Shing Leu:** Data curation, Investigation, Methodology, Writing – review & editing. **Fang-Ju Sun:** Formal analysis, Methodology, Visualization. **Yu-Jen Chen:** Data curation, Investigation, Methodology, Writing – review & editing. **Jie Lee:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Methodology, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

Ethics approval: This study was conducted in accordance with the principles of the Declaration of Helsinki. This study was approved by the Institutional Review Boards of MacKay Memorial Hospital (23MMHIS437e) and Changhua Christian Hospital (IRB No. 180503).

Author contribution

Study concept and design: Jie Lee.
Collection and assembly of data: Jhen-Bin Lin, Yu-Jen Chen, and Yi-Shing Leu.
Data analysis and interpretation: Yu-Fan Wu, Fang-Ju Sun, and Jie Lee.
Manuscript writing: Yu-Fan Wu and Jie Lee.
Critical revision of the manuscript for important intellectual content: All authors.
Final approval of manuscript: All authors.

Funding/support

This work was supported by the National Science and Technology Council of Taiwan (grant number Contract No. MOST 110-2314-B-195-033, NSTC 113-2314-B-195-011-MY3, and NSTC 113-2221-E-A49-167-MY3), and MacKay Memorial Hospital (grant number: MMH-114-22).

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.radonc.2025.111066>.

Data availability

The datasets used and/or analyzed in the current study are available from the corresponding author upon reasonable request.

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